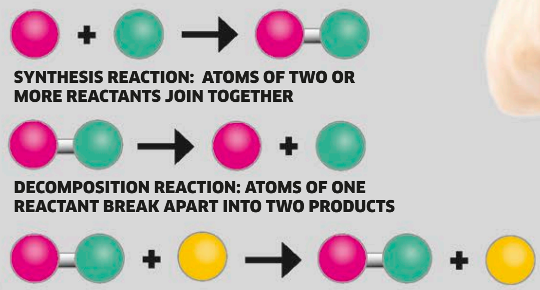
CHEMICAL REACTIONS

A chemical reaction is what happens when one substance meets and reacts with another and a new substance is formed. The substances that react together are called reactants, and those formed are called products. In a chemical reaction, atoms are only rearranged, never created or destroyed.

A change in color of a substance often indicates that a reaction has happened.

DIFFERENT REACTIONS

There are many types of reaction. They vary depending on the reactants involved and the conditions in which they take place. Some reactions happen in an instant, and some take years. Exothermic reactions give off heat while endothermic reactions cool things down. The products in a reversible reaction can turn back into the reactants, but in an irreversible reaction they cannot. Redox reactions involve two simultaneous reactions: reduction and oxidation.



SYNTHESIS REACTION: ATOMS OF TWO OR MORE REACTANTS JOIN TOGETHER

DECOMPOSITION REACTION: ATOMS OF ONE REACTANT BREAK APART INTO TWO PRODUCTS

DISPLACEMENT REACTION: ATOMS OF ONE TYPE SWAP PLACES WITH THOSE OF ANOTHER, FORMING NEW COMPOUNDS

Three kinds of reaction

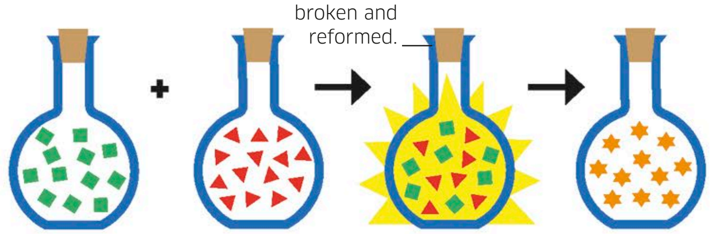
Reactions can be classified in three main groups according to the fate of the reactants. As shown above, in some reactions the reactants join together, in others they break apart, and in some their atoms swap places.

REACTION BASICS

Chemical reactions are going on around us all the time. They help us digest food, they cause metal to rust, wood to burn, and food to rot. Chemical reactions can be fun to watch in a laboratory—they can send sparks flying, create puffs of smoke, or trigger dramatic color changes. Some happen quietly, however, without us even noticing. The important fact behind all these reactions is that all the atoms involved remain unchanged. The atoms that were there at the beginning of the reactions are the same as the atoms at the end of the reaction. The only thing that has changed is how those atoms have been rearranged.

Reactants and products

The result of a chemical reaction is a chemical change, and the generation of a product or products that are different from the reactants. Often, the product looks nothing like the reactants. A solid might be formed by two liquids, a yellow liquid might turn blue, or a gas might be formed when a solid is mixed with a liquid. It doesn't always seem as if the atoms in the reactants are the same as those in the products, but they are.

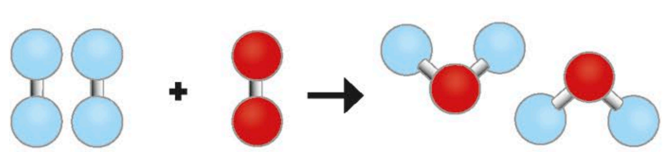


Bonds are broken and reformed.

REACTANT 1 | REACTANT 2 | REACTION | PRODUCT

Chemical equations

The "law of conservation of mass" states that mass is neither created nor destroyed. This applies to the mass of the atoms involved in a reaction, and can be shown in a chemical equation. Reactants are written on the left, and products on the right. The number of atoms on the left of the arrow always equal those on the right. Everything is abbreviated: "2 H_2" means two molecules of hydrogen, with two atoms in each molecule.



2 HYDROGEN MOLECULES ($2 H_2$) | 1 OXYGEN MOLECULE (O_2) | 2 WATER MOLECULES ($2 H_2O$)

Dirty exhaust in

A car's catalytic converter contains a catalyst made of platinum and rhodium.

Carbon monoxide and unburned fuel are converted to harmless carbon dioxide and water as they pass through the converter.

Cleaner exhaust out

Catalysts

Catalysts are substances that make chemical reactions go faster. Some reactions can't start without a catalyst. Catalysts help reactants interact, but they are not part of the reaction and remain unchanged. Different catalysts do different jobs. Cars use catalysts that help reduce harmful engine fumes by speeding up their conversion to cleaner exhausts.